Advait Deshmukh

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Research Interests

Natural Language Processing, Narrative Understanding, Computational Discourse, Interpretability

Education

Doctor of Philosophy, Computer Science — *Jan 2026 – Dec 2030*
University of Colorado Boulder — Boulder, CO
Advised by Dr. Maria Antoniak

Master of Science, Computer Science — *Aug 2023 – Dec 2025*
University of Colorado Boulder — Boulder, CO
Advised by Dr. Maria Leonor Pacheco

Bachelor of Engineering, Computer Science — *Aug 2017 – Aug 2021*
Birla Institute of Technology and Science, Pilani — Pilani, India

Publications

[1] Narrative-UFET: Narrative Generation for Ultra-Fine Entity Typing.
Mreedul Gupta, **Advait Deshmukh**, Ashwin Umadi, Matt Pauk, and Maria Leonor Pacheco. 2027.
Proceedings of the 20th Conference of the European Chapter of the Association for Computational Linguistics (EACL 2027), Athens, Greece.

[2] The Garden of Forking Prompts: How Users Explore Narrative Space in Story Generation.
Advait Deshmukh, Nora Benedict, Melanie Walsh, and Maria Antoniak. 2026.
Proceedings of the Conference on Language Modeling (COLM 2026), San Francisco, CA.

[3] A Structured Clustering Approach for Inducing Media Narratives.
Rohan Das, **Advait Deshmukh**, Alexandria Leto, Zohar Naaman, I-Ta Lee, and Maria Leonor Pacheco. 2026.
Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (ACL 2026), San Diego, CA.

[4] All Entities are Not Created Equal: Examining the Long Tail for Ultra-Fine Entity Typing.
Advait Deshmukh, Ashwin Umadi, Dananjay Srinivas, and Maria Leonor Pacheco. 2025.
*Proceedings of the 14th Joint Conference on Lexical and Computational Semantics (*SEM 2025)*, co-located with EMNLP 2025.

Non-Archival Presentations

[5] The Garden of Forking Prompts: How Users Explore Narrative Space in Story Generation.
Advait Deshmukh, Nora Benedict, Melanie Walsh, and Maria Antoniak. 2026.
New Directions in Analyzing Text as Data (TADA 2026), Berkeley, CA.

Selected Research Projects

How Users Explore Narrative Space in Story Generation — *Spring 2026 – Summer 2026*
Advised by Dr. Maria Antoniak — CLS Lab, CU Boulder

- Constructed WildStories, a sample of 275,635 story generation prompts, and WildEdits, 24,291 edit trees tracing how users iteratively revise prompts, from naturally occurring user-chatbot conversations in WildChat.
- Developed a framework of edit types crossing four directions with fourteen narrative targets, and showed that automated permutations of it can be used to benchmark story generation.

A Structured Clustering Approach for Inducing Media Narratives — *Spring 2025 – Summer 2025*
Collaborator — BLAST Lab, CU Boulder

- Implemented a modified DAPrompt (Xiang et al.) based relationship classifier to extract causally related event pairs across 4 datasets.
- Infused knowledge from teacher models (Llama3, Qwen3) to aid in domain adaptation.

Examining the Long Tail for Ultra-Fine Entity Typing — *Fall 2024 – Spring 2025*
Independent study advised by Dr. Maria Pacheco — BLAST Lab, CU Boulder

- Conducted rigorous empirical analysis to study limitations of predominantly popular language model-based approaches for ultra-fine entity typing.
- Developed proxies to predict entity frequencies based on pretrained language models and analyzed its effect on typing performance.

Teaching Experience

University of Colorado Boulder *Fall 2024 – Spring 2025*
Teaching Assistant, CSCI 1300: Intro to Computing Boulder, CO

- Assisted in the course by grading, holding recitations & office hours to support students.
- Collaborated with faculty to redesign course materials, improving student engagement.

University of Colorado Boulder *Fall 2023 – Spring 2024*
Course Manager, CSCI 1300: Intro to Computing Boulder, CO

- Managed operations for the course, serving over 900 students through office hours and online platform Ed.
- Collaborated with faculty to create and grade assignments, projects & exams.

Professional Experience

Amazon *Summer 2026*
Applied Science Intern Bellevue, WA

- Developed an atomic claim evaluation (ACE) system to evaluate agentic systems for the Supply Chain Optimization Technologies (SCOT) long-term planning & forecasting team.
- Improved error detection precision by 30% and recall by 50% over an LLM-as-judge baseline, across both offline evaluation against human outputs and live evaluation against source documents.

Societe Generale Global Solution Centre *Jul 2021 – Jul 2023*
Software Development Engineer Bangalore, India

- Developed a Retrieval Augmented Generation (RAG) based chat system to assist support teams in efficiently handling policy queries.
- Built an innovation portal, generating 120 ideas across 7 departments to foster cross-departmental collaboration.

Centre for Development of Advanced Computing (CDAC) *Jul 2020 – Dec 2020*
Research Intern Pune, India

- Engineered a scalable platform for High Performance Computing (HPC) job submission to optimize data processing.
- Developed a portal for seamless management of big data tasks, enhancing research efficiency.

VoiceQube *May 2019 – Jul 2019*
Summer Intern Bangalore, India

- Participated in summer internship program as part of the practice school program at BITS Pilani.
- Developed a conversational service to assist students in their GRE preparation.

Skills

Programming Languages: Python, JavaScript, TypeScript, C++, Java
Frameworks & Tools: React, Spring, FastAPI, Flask
Machine Learning Libraries: PyTorch, Transformers, NLTK, Pandas, Numpy, Langchain, Ollama
Cloud Technologies: Google Cloud, AWS, Microsoft Azure
DevOps & Version Control: Docker, Git, Kubernetes

Awards and Grants

Conference Support Funds – $1500 *Dec 2025*
Computer Science Department, University of Colorado Boulder, USA

Travel Subsidy Recipient – $750 *Nov 2025*
**SEM 2025, co-located with EMNLP 2025* Suzhou, China

Lloyd Botway Fellow – $500 *April 2025*
Computer Science Department, University of Colorado Boulder, USA

Outstanding Teaching Assistant – $500 *March 2025*
CSCI 1300, University of Colorado Boulder, USA

Winner - Poster Competition – $500 *Feb 2025*
Annual Research Expo, University of Colorado Boulder, USA

Grad App Aid Recipient – $500 *Nov 2024*
QueerInAI Boulder, USA

Leadership Experience

Founder and President *Sep 2019 – May 2021*
VoiceQube Pilani

- Founded and led VoiceQube BITS Pilani Chapter to develop Voice interface-based projects.
- Deployed BITS Buddy to handle queries about campus life, responding to ∼4000 requests between January and March 2020.